1D FEM elements

1 Geometry

The following terms are defined:

n_{el} : Total number of elements.
n_{ne} : Number of nodes per element.
n : Total number of nodes ($n = n_{ne} + (n_{el} - 1)(n_{ne} - 1)$).
n_G : Number of Gauss points.

The following matrices need to be defined:

$[\mathbf{x}]$: Nodal coordinates array ($n \times 1$).

$$[\mathbf{x}] = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$[\mathbf{T}_n]$: Nodal connectivity table ($n_{el} \times n_{ne}$).

For each element $e = 1 \dots n_{el}$

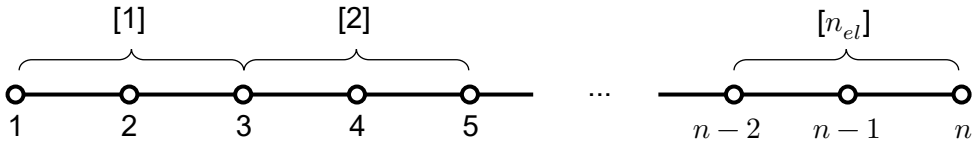
For each element node $i = 1 \dots n_{ne}$

$$\mathbf{T}_n(e, i) = (e - 1)(n_{ne} - 1) + i$$

Next node i

Next element e

Example ($n_{ne} = 3$):



2 Computation of the global stiffness matrix

Initialize matrices:

$[\mathbf{B}] = [\mathbf{0}]_{1 \times n_{ne} \times n_{el} \times n_G}$

$[\mathbf{K}^e] = [\mathbf{0}]_{n_{ne} \times n_{ne} \times n_{el}}$

$[\mathbf{K}] = [\mathbf{0}]_{n \times n}$

For each element $e = 1 \dots n_{el}$

a) Get element's size

$$l^e = \mathbf{x}\big(\mathbf{T}_n(e, n_{ne})\big) - \mathbf{x}\big(\mathbf{T}_n(e, 1)\big)$$

For each Gauss point $k = 1 \dots n_G$ (see Appendix B)

b) Compute $\mathbf{B}$ *matrix evaluated at* ξ_k *(see Appendix A)*

For each element node $i = 1 \dots n_{ne}$

$$\mathbf{B}(1, i, e, k) = \frac{1}{J^e(\xi_k)} \frac{\partial N_i^e}{\partial \xi}(\xi_k)$$

Next node i

c) Compute the element stiffness matrix evaluated at ξ_k

$$\mathbf{K}^e(:,:,e) = \mathbf{K}^e(:,:,e) + w_k J^e(\xi_k) E^e A^e [\mathbf{B}(:,:,e,k)]^{\mathrm{T}} [\mathbf{B}(:,:,e,k)]$$

Next Gauss point k

d) Assembly to global stiffness matrix

For each element node $i = 1 \dots n_{ne}$

For each element node $j = 1 \dots n_{ne}$

$$\mathbf{K}\big(\mathbf{T}_n(e,i), \mathbf{T}_n(e,j)\big) = \mathbf{K}\big(\mathbf{T}_n(e,i), \mathbf{T}_n(e,j)\big) + \mathbf{K}^e(i,j,e)$$

Next node j

Next node i

Next element e

3 Computation of the global force vector

Initialize matrices:

$[\mathbf{N}] = [\mathbf{0}]_{1 \times n_{ne} \times n_{el} \times n_G}$

$[\mathbf{x}_G] = [\mathbf{0}]_{n_G \times n_{el}}$

$[\mathbf{f}^e] = [\mathbf{0}]_{n_{ne} \times n_{el}}$

$[\mathbf{f}] = [\mathbf{0}]_{n \times 1}$

For each element $e = 1 \dots n_{el}$

a) Get nodal coordinates and element's size

$$\mathbf{x}^e = \mathbf{x}\big(\mathbf{T}_n(e,:)\big)$$

$$l^e = \mathbf{x}\big(\mathbf{T}_n(e, n_{ne})\big) - \mathbf{x}\big(\mathbf{T}_n(e,1)\big)$$

For each Gauss point $k = 1 \dots n_G$ (see Appendix B)

b) Compute $\mathbf{N}$ *matrix evaluated at* ξ_k *(see Appendix A)*

For each element node $i = 1 \dots n_{ne}$

$$\mathbf{N}(1,i,e,k) = N_i^e(\xi_k)$$

Next node i

c) Compute Gauss point coordinate and element force vector evaluated at ξ_k

$$\mathbf{x}_G(k,e) = [\mathbf{N}(:,:,e,k)]\{\mathbf{x}^e\}$$

d) Evaluate body force at Gauss point

$$b_k = b\big(\mathbf{x}_G(k,e)\big)$$

e) Compute element force vector evaluated at ξ_k

$$\mathbf{f}^e(:,e) = \mathbf{f}^e(:,e) + w_k J^e(\xi_k)[\mathbf{N}(:,:,e,k)]^{\mathrm{T}} b_k$$

Next Gauss point k

d) Assembly to global force matrix

For each element's node $i = 1 \dots n_{ne}$

$$\mathbf{f}(\mathbf{T}_n(e,i),1) = \mathbf{f}(\mathbf{T}_n(e,i),1) + \mathbf{f}^e(i,e)$$

Next node i

Next element e

4 Global system of equations

a) Apply conditions

$\nu_r = [\dots]$: Array with the imposed degrees of freedom.

$\nu_f = [\dots]$: Array with the free degrees of freedom.

b) Partitioned system of equations

$$\begin{bmatrix}\mathbf{K}_{rr} & \mathbf{K}_{rf}\\ \mathbf{K}_{fr} & \mathbf{K}_{ff}\end{bmatrix}\begin{Bmatrix}\mathbf{u}_r\\ \mathbf{u}_f\end{Bmatrix} = \begin{Bmatrix}\mathbf{f}_r\\ \mathbf{f}_f\end{Bmatrix}$$

where

$$\mathbf{K}_{ff} = \mathbf{K}(\nu_f,\nu_f)$$
$$\mathbf{K}_{fr} = \mathbf{K}(\nu_f,\nu_r)$$
$$\mathbf{K}_{rf} = \mathbf{K}(\nu_r,\nu_f)$$
$$\mathbf{K}_{rr} = \mathbf{K}(\nu_r,\nu_r)$$

Tip: In Matlab, ν_f can be obtained automatically once ν_r and the total number of degrees of freedom n have been defined with the command:

```
vf = setdiff(1:n,vr)
```

c) System resolution

$$\mathbf{u}(\nu_r,1) = \mathbf{u}_r$$
$$\mathbf{u}(\nu_f,1) = (\mathbf{K}_{ff})^{-1}\left(\mathbf{f}(\nu_f,1) - \mathbf{K}_{fr}\mathbf{u}(\nu_r,1)\right)$$
$$\mathbf{f}_r = \mathbf{K}_{rr}\mathbf{u}(\nu_r,1) + \mathbf{K}_{rf}\mathbf{u}(\nu_f,1) - \mathbf{f}(\nu_r,1)$$

5 Computation of stress

Initialization:

$[\boldsymbol{\sigma}] = [\mathbf{0}]_{n_G \times n_{el}}$

For each element $e = 1 \dots n_{el}$

a) Obtain element nodes displacement

$\mathbf{u}^e = \mathbf{u}\big(\mathbf{T}_n(e,:)\big)$

b) Obtain stress at Gauss point

For each Gauss point $k = 1 \dots n_G$ (see Appendix B)

$\boldsymbol{\sigma}_G(k,e) = E[\mathbf{B}(:,:,e,k)]\{\mathbf{u}^e\}$

Next Gauss point k

Next element e

Tip: From the stress, `sigG`, at each Gauss point, `xG`, one can easily interpolate the corresponding value at the nodes using Matlab's function:

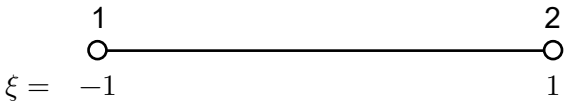
```
sig = interp1(xG(:),sigG(:),x)
```

Then, the plot of the stress distribution is realized using:

```
plot(x,sig)
```

Appendix A. Elements and shape functions

2-noded linear element:

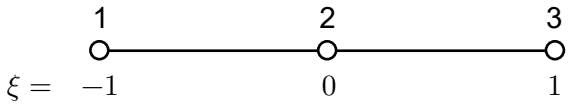


$$\hat{N}_1(\xi) = \frac{1}{2}(1-\xi); \qquad \hat{N}_2(\xi) = \frac{1}{2}(1+\xi)$$

$$\frac{d\hat{N}_1}{d\xi}(\xi) = -\frac{1}{2}; \qquad \frac{d\hat{N}_2}{d\xi}(\xi) = +\frac{1}{2}$$

$$J^e(\xi) = \frac{d\hat{N}_1}{d\xi}x_1^e + \frac{d\hat{N}_2}{d\xi}x_2^e$$

3-noded quadratic element:



$$\hat{N}_1(\xi) = \frac{1}{2}\xi(\xi-1); \qquad \hat{N}_2(\xi) = (1-\xi^2); \qquad \hat{N}_3(\xi) = \frac{1}{2}\xi(\xi+1)$$

$$\frac{d\hat{N}_1}{d\xi}(\xi) = \xi - \frac{1}{2}; \qquad \frac{d\hat{N}_2}{d\xi}(\xi) = -2\xi; \qquad \frac{d\hat{N}_3}{d\xi}(\xi) = \xi + \frac{1}{2}$$

$$J^e(\xi) = \frac{d\hat{N}_1}{d\xi}x_1^e + \frac{d\hat{N}_2}{d\xi}x_2^e + \frac{d\hat{N}_3}{d\xi}x_3^e$$

Appendix B. Gauss quadrature points and weights

n_G	w_k	ξ_k
1	2	0
2	1	$-1/\sqrt{3}$
		$+1/\sqrt{3}$
3	$^8/_9$	0
	$^5/_9$	$+\sqrt{3/5}$
		$-\sqrt{3/5}$